Sliding to rolling

LAMMPS dem05 | Tangential friction and sphere rotation | 28 October 2026

FAIL	0 of 4 runs pass both official 1% checks; the benchmark is not validated.

Official endpoint check at 0.75 s

Run	dt (us)	vx (m/s)	Error (%)	omega_y (rad/s)	Error (%)	1% checks
CPU dt	10	0.70348380	1.51227	289.321928	1.26267	FAIL
GPU dt	10	0.70348380	1.51227	289.321928	1.26267	FAIL
CPU dt/2	5	0.70348380	1.51227	289.321920	1.26267	FAIL
GPU dt/2	5	0.70348380	1.51227	289.321920	1.26267	FAIL

Targets: vx = 5/7 = 0.714285714286 m/s; omega_y = vx/R = 285.714285714 rad/s. The official slip_cessation check requires each relative error to be at most 1%. The same inputs and criterion were used for all four runs.

Inputs and exact material mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	Diameter 0.005 m; density 2700 kg/m^3; center (0, 0, 0.0024393) m; vx = 1 m/s; zero spin
Wall / gravity	Floor z = 0; upper plane z = 1 (inactive); g = 1 m/s^2 downward
Contact materials	Both E = 1e4 Pa and Poisson ratio = 0.33; E* = 5611.04253 Pa; G* = 1125.56841 Pa
Contact / damping	Hertz-Mindlin; sliding friction 0.4; e = 1; automatic contact damping = 0; rolling, torsion and global damping = 0
Time / retained output	0.75 s; dt = 10 / 5 us; 75,000 / 150,000 steps; 751 native frames per run at 1 ms; every-step CSV retained

Ideal sliding and rolling

I = (2/5) m R^2; vx = 1 - 0.4 t; omega_y = 400 t.
s_R = vx - R omega_y = 1 - 1.4 t; ts = 0.714285714286 s.
After ideal slip cessation: vx = 5/7 and omega_y = (5/7)/R; final kinetic-energy fraction = 5/7. The official check does not directly test ts.

Contact geometry explains a different target

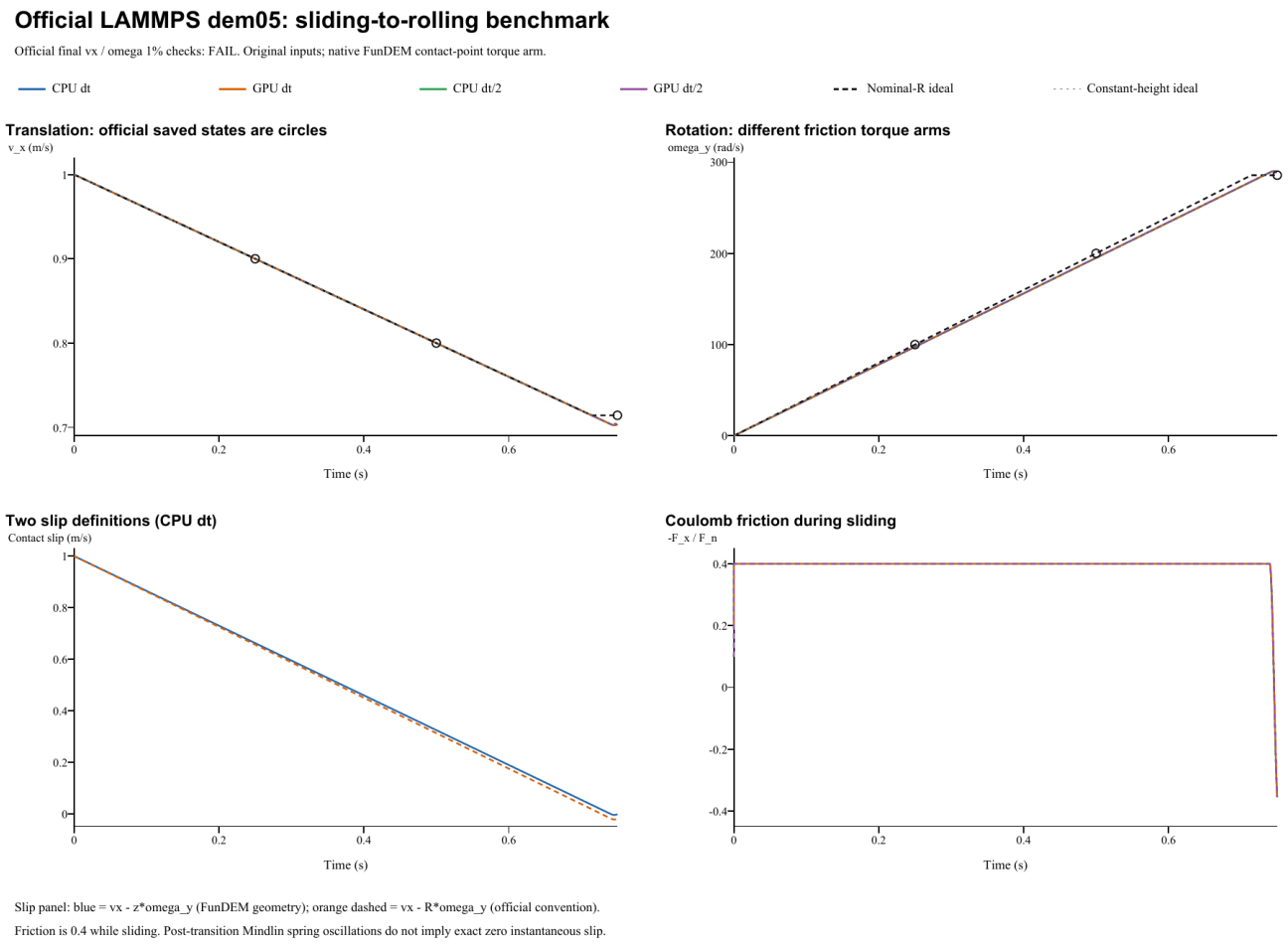
FunDEM uses a = R - delta = z for slip and torque; LAMMPS uses R. Initial overlap is 60.7 um (2.4280% of R). A constant-height model at h = 0.0024393431 m predicts vx = 0.704155836 m/s, omega_y = 288.666170 rad/s, ts = 0.739610409 s. This explains the convention, not a replacement acceptance target or a demonstrated solver bug.

Reference provenance

Official LAMMPS YAML and golden states; analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML Git blob 864e4de3731c20e3ee0941e946439adfd40bfbbe. LAMMPS was not rerun locally. The YAML epsilon of 1e-10 is not a universal cross-code tolerance. Periodic x length = 0.55 m; position comparisons account for wrapping.

Curves and retained evidence

Native trajectories, official checkpoints and the ideal nominal-radius solution



Nominal slip s_R = vx - R omega_y; geometric slip s_a = vx - z omega_y. Mean sliding abs(Ft)/Fn ranges from 0.4000000 to 0.4000000. Maximum torque-derived lever-arm error = 4.34e-19 m. Final base-step nominal / geometric slips are -0.01982103 / -0.002270022 m/s. Sustained exact pure rolling is not demonstrated.

Complete native results are retained

CPU dt: run-20261028-042729-846-7c302c7f-935d-419c-b858-1dc5c539b072; 75,000 steps / 751 native frames.

GPU dt: run-20261028-042730-008-1cbf0d7d-ece8-42c7-bcbc-737265403759; 75,000 steps / 751 native frames.

CPU dt/2: run-20261028-042846-183-68d500ee-dcce-427e-9279-153629dde3e0; 150,000 steps / 751 native frames.

GPU dt/2: run-20261028-043017-288-e3f3e0be-7bc6-4272-8bdd-6cff3e1eac95; 150,000 steps / 751 native frames.

Run manifest | Independent analysis JSON | Checkpoint CSV | Contact-model audit. Native run.json, project.json, configured frames/contacts, projects, audits, backend logs and per-step CSVs are retained.

Native output is every 1 ms, including both endpoints: 751 frames per run, every 100 base steps or 200 half steps. All configured output frames are retained; every integration step is separately recorded in the trajectory CSV. A native frame was not written at every step.

Execution and scope

Real FunDEM SimulationSession runs used linked 0.4.3 CPU/CUDA native libraries. Maximum CPU/GPU differences: 3.33e-15 m/s and 5.74e-12 rad/s. Halving dt changes final vx by 8.4e-09 m/s and omega_y by 8.2e-06 rad/s. No solver changes. GUI byte identity was not established.

This case exercises sliding friction, spin-up and transition toward rolling. CPU/GPU agreement or dt refinement cannot override a failed official reference check. Rolling resistance and torsional friction are zero; nonzero tangential damping and many-particle flow require other tests.